

SIMATS ENGINEERING ASSESSMENT TOOLS

COURSE NAME: Theory of Computation

COURSE CODE: CSA1304

COURSE OUTCOME COVERED

CO4: Construct Context-Free Grammars, generate derivations and parse trees to demonstrate the syntactic structure of strings. (BL : 3)

Assessment Tool Weightage: 10%

QUESTIONS: 10 Total Marks:100 Data Interpretation

Code of Conduct:

I M.VINAY(192311146) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

M.VINAY

Signature of the student:

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Given Data (Grammar)

Consider the CFG:

$S \rightarrow aSb \mid ab$

Q1. Language Identification

Based on the given grammar, which of the following strings belong to the language? A) ab

B) aabb

C) aaabbb

D) abb

Q2. Pattern Recognition

From the grammar, identify the correct pattern:

A) $a^n b^n$

B) $(ab)^n$

C) $a^n b^m$

D) $(a+b)^*$

Q3. Derivation Analysis

Which of the following is a valid derivation for aabb?

A) $S \rightarrow aSb \rightarrow aaSbb \rightarrow aabb$

B) $S \rightarrow ab \rightarrow aabb$

C) $S \rightarrow aSb \rightarrow ab$

D) $S \rightarrow aaSbb \rightarrow aabb$

Q4. Missing Step Identification

Fill in the missing step:

$S \rightarrow aSb \rightarrow aaSbb \rightarrow \underline{\hspace{2cm}} \rightarrow aabb$

A) ab

B) S

C) ϵ

D) aSb

Q5. Parse Tree Interpretation

How many levels are there in the parse tree for string aaabbb?

A) 2

B) 3

C) 4

D) 5

Answer: D

Consider:

$S \rightarrow SS \mid a$

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Q6. Ambiguity Check

This grammar is:

- A) Unambiguous
- B) Ambiguous
- C) Regular
- D) Deterministic

Q7. String Generation

Which string can be generated?

- A) a
- B) aa
- C) aaa
- D) All of the above

Q8. Interpretation of Ambiguity

Ambiguity means:

- A) One string $\rightarrow$ one parse tree
- B) One string $\rightarrow$ multiple parse trees
- C) No derivation
- D) Infinite symbols

Given Data (Grammar 1)

$S \rightarrow 0S1 \mid 01$

Q9. Derivation Interpretation

Which of the following derivations correctly generates the string

0011? A) $S \rightarrow 0S1 \rightarrow 00S11 \rightarrow 0011$

B) $S \rightarrow 01 \rightarrow 0011$

C) $S \rightarrow 0S1 \rightarrow 01$

D) $S \rightarrow 00S11 \rightarrow 0011$

Given Data (Grammar 2)

$S \rightarrow aS \mid Sb \mid ab$

Q10. Structural Analysis

Which of the following strings can be generated by the grammar? A) ab

B) aab

C) abb

D) All of the above

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RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

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QUESTION 1

Given CFG

$$S \rightarrow aSb \mid ab$$

Question

Which of the following strings belongs to the language generated by the CFG?

A) aabb

B) aaabbbb

C) abab

D) aababb

Answer

A) aabb

Derivation

Start with:

S

Apply:

$$S \rightarrow aSb$$

Therefore:

$$S \Rightarrow aSb$$

Again apply:

$$S \rightarrow aSb$$

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Therefore:

$$aSb \Rightarrow aaSbb$$

Now apply the terminating production:

$$S \rightarrow ab$$

Therefore:

$$aaSbb \Rightarrow aa(ab)bb$$

Hence:

$$S \Rightarrow aSb \Rightarrow aaSbb \Rightarrow aabb$$

Explanation

The production

$$S \rightarrow aSb$$

adds one a at the beginning and one b at the end.

The production

$$S \rightarrow ab$$

terminates the derivation.

For aabb:

$$aabb = a^2b^2$$

Thus, the string follows the pattern generated by the CFG.

Final Answer

A) aabb

QUESTION 2

Pattern Recognition

Question

Consider the CFG:

$$S \rightarrow aSb \mid ab$$

Identify the correct pattern generated by the grammar.

A) $a^n b^n$

B) $(ab)^n$

C) $a^n b^m$

D) $(a+b)^*$

Answer

A) $a^n b^n$

1. Given Grammar

$$S \rightarrow aSb \mid ab$$

There are two production rules:

$$S \rightarrow aSb$$

And

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$$S \rightarrow ab$$

2. Understanding the Production

The rule

$$S \rightarrow aSb$$

adds:

- one a to the left side
- one b to the right side

Therefore, every time this rule is applied, the number of as and bs increases by one.

The derivation finally stops using:

$$S \rightarrow ab$$

Thus, the number of as is always equal to the number of bs.

3. Derivation for

Using:

$$S \rightarrow ab$$

we get:

$$S \Rightarrow ab$$

Therefore:

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a1b1

4. Derivation for

Start with:

S

Apply:

$S \rightarrow aSb$

Therefore:

$S \Rightarrow aSb$

Now apply:

$S \rightarrow ab$

Therefore:

$aSb \Rightarrow a(ab)b$

Hence:

$S \Rightarrow aSb \Rightarrow aabb$

This gives:

a2b2

5. Derivation for aaabbb

Start with:

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S

Apply:

$$S \rightarrow aSb$$

$$S \Rightarrow aSb$$

Apply the same rule again:

$$aSb \Rightarrow aaSbb$$

Now use:

$$S \rightarrow ab$$

Therefore:

$$aaSbb \Rightarrow aa(ab)bb$$

Hence:

$$S \Rightarrow aSb \Rightarrow aaSbb \Rightarrow aaabbb$$

This gives:

$$a^3b^3$$

6. General Derivation

The grammar generates:

$$S \Rightarrow aSb$$

$$\Rightarrow aaSbb$$

$$\Rightarrow \dots$$

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Finally:

$$S \Rightarrow ab$$

Therefore:

$$a^n b^n, n \geq 1$$

7. Pattern Analysis

The generated strings are:

ab

aabb

aaabbb

aaaabbbb

and so on.

Observe:

ab $\rightarrow$ 1 a, 1 b

aabb $\rightarrow$ 2 a, 2 b

aaabbb $\rightarrow$ 3 a, 3 b

aaaabbbb $\rightarrow$ 4 a, 4 b

In every case:

$$\text{Number of a's} = \text{Number of b's}$$

and all as occur before all bs.

8. Why the Other Patterns Are Not Correct

Option B:

This produces:

ab, abab, ababab,...

But the CFG produces:

ab, aabb, aaabbb,...

Therefore, this is not the correct pattern.

Option C:

This allows a and b to have different numbers.

But the CFG always produces equal numbers.

Therefore, it is not the correct pattern.

Option D:

This represents arbitrary strings containing a and b.

The CFG does not generate arbitrary strings.

Therefore, it is not the correct pattern.

9. Final Result

The CFG

$$S \rightarrow aSb \mid ab$$

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generates strings having equal numbers of as and bs, with all as appearing before all bs.

Therefore, the language is:

$$L = \{anbn | n \geq 1\}$$

Hence, the correct answer is:

A) $anbn$

QUESTION 3

Derivation Analysis

Question

Which of the following is a valid derivation for aabb?

A) $S \rightarrow aSb \rightarrow aaSbb \rightarrow aabb$

B) $S \rightarrow ab \rightarrow aabb$

C) $S \rightarrow aSb \rightarrow ab$

D) $S \rightarrow aaSbb \rightarrow aabb$

Answer

A) $S \rightarrow aSb \rightarrow aaSbb \rightarrow aabb$

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Given CFG

$$S \rightarrow aSb \mid ab$$

Step-by-Step Derivation

Start with:

S

Apply the production:

$$S \rightarrow aSb$$

Therefore:

$$S \rightarrow aSb$$

Apply the same production again:

$$aSb \rightarrow aaSbb$$

Now replace S using:

$$S \rightarrow ab$$

Therefore:

$$aaSbb \rightarrow aa(ab)bb$$

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Simplifying:

$aa(ab)bb \rightarrow aabb$

Complete Derivation

S

$\rightarrow aSb$

$\rightarrow aaSbb$

$\rightarrow aa(ab)bb$

$\rightarrow aabb$

Verification

The final string is:

aabb

It contains:

2 a's

2 b's

and follows the required order:

aa followed by bb

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Therefore, the derivation is valid.

Why Option A Is Correct

The derivation starts from the start symbol S, applies valid grammar productions at every step, and finally produces the required string aabb.

$$S \rightarrow aSb$$

is valid.

$$aSb \rightarrow aaSbb$$

is valid.

$$aaSbb \rightarrow aa(ab)bb$$

is valid because:

$$S \rightarrow ab$$

Hence:

$$aa(ab)bb \rightarrow aabb$$

is the required final string.

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Final Answer

A) $S \rightarrow aSb \rightarrow aaSbb \rightarrow aabb$

QUESTION 4

Missing Step Identification

Question

Fill in the missing step:

$S \rightarrow aSb \rightarrow aaSbb \rightarrow \underline{\hspace{2cm}} \rightarrow aabb$

Options:

A) ab

B) S

C) ε

D) aSb

The question and options are given exactly in the assessment document.

Answer

C) ε

Given CFG

$S \rightarrow aSb \mid ab$

Step-by-Step Derivation

Start with:

S

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Apply:

$$S \rightarrow aSb$$

Therefore:

$$S \rightarrow aSb$$

Apply the same production again:

$$aSb \rightarrow aaSbb$$

Now the required next step in the question is:

$$aaSbb \rightarrow aa\epsilon bb$$

Since ϵ represents the empty string:

$$aa\epsilon bb \rightarrow aabb$$

Complete Derivation

S

$$\rightarrow aSb$$

$$\rightarrow aaSbb$$

$$\rightarrow aa\epsilon bb$$

$$\rightarrow aabb$$

Explanation

The symbol ϵ represents an empty string. When S is replaced by ϵ , nothing is added between the two a s and two b s:

$$aa\epsilon bb = aabb$$

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Therefore, the missing step shown in the question is:

ε

Final Answer

C) ε

QUESTION 5

Question

How many levels are there in the parse tree for the string aaabbb?

A) 2

B) 3

C) 4

D) 5

Given CFG

$S \rightarrow aSb \mid ab$

Answer

D) 5

Derivation

Starting from the start symbol:

S

$\rightarrow aSb$

$\rightarrow aaSbb$

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→ aaaSbbb

→ aaabbb

The final string obtained is:

aaabbb

Parse Tree Levels

The derivation can be represented level by level:

Level 1: S

Level 2: a S b

Level 3: a a S b b

Level 4: a a a S b b b

Level 5: a a a b b b

Therefore, the parse tree contains:

5 levels

Explanation

The production

$S \rightarrow aSb$

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adds one a to the left and one b to the right at each step.

For aaabbb, the derivation requires three applications of this structure before replacing S with the final ab.

The complete derivation is:

S

→ aSb

→ aaSbb

→ aaaSbbb

→ aaabbb

Hence, the parse tree has **5 levels** when the root and final terminal level are included.

Final Answer

D) 5

QUESTION 6

Question

Given the CFG:

$S \rightarrow SS \mid a$

Which of the following strings can be generated by the grammar?

A) aa

B) aaa

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C) aaaa

D) All of the above

Given CFG

$S \rightarrow SS \mid a$

Answer

D) All of the above

Explanation

The grammar has two productions:

$S \rightarrow SS$

$S \rightarrow a$

The production:

$S \rightarrow a$

produces a single a.

The production:

$S \rightarrow SS$

combines two strings generated by S.

Therefore, the grammar can generate strings containing one or more as.

Derivation for aa

S

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$\rightarrow SS$

$\rightarrow aS$

$\rightarrow aa$

Therefore:

$aa \in L$

Derivation for aaa

S

$\rightarrow SS$

$\rightarrow aS$

$\rightarrow aSS$

$\rightarrow aaS$

$\rightarrow aaa$

Therefore:

$aaa \in L$

Derivation for aaaa

S

$\rightarrow SS$

$\rightarrow aS$

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→ aSS

→ aaS

→ aaSS

→ aaaS

→ aaaa

Therefore:

aaaa $\in$ L

General Pattern

The grammar can generate:

a

aa

aaa

aaaa

aaaaa

...

Therefore, the language is:

$$L = \{a^n \mid n \geq 1\}$$

Final Answer

D) All of the above

QUESTION 7

Question

Consider the CFG:

$$S \rightarrow SS \mid a$$

Which string can be generated?

A) a

B) aa

C) aaa

D) All of the above

Answer

D) All of the above

Given CFG

$$S \rightarrow SS \mid a$$

The grammar has two productions:

$$S \rightarrow SS$$

$$S \rightarrow a$$

The production $S \rightarrow a$ generates a single a.

The production $S \rightarrow SS$ combines two strings generated from S.

Derivation for a

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S

$\rightarrow a$

Therefore:

$a \in L$

Derivation for aa

S

$\rightarrow SS$

$\rightarrow aS$

$\rightarrow aa$

Therefore:

$aa \in L$

Derivation for aaa

S

$\rightarrow SS$

$\rightarrow aS$

$\rightarrow aSS$

$\rightarrow aaS$

$\rightarrow aaa$

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Therefore:

$aaa \in L$

Explanation

The grammar can generate one or more as.

The production:

$S \rightarrow a$

provides the basic symbol a.

The production:

$S \rightarrow SS$

allows two generated strings to be combined.

Therefore, the grammar generates:

a

aa

aaa

aaaa

aaaaa

...

Hence, all three given strings can be generated.

Final Answer

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D) All of the above

QUESTION 8

Question

Consider the CFG:

$$S \rightarrow SS \mid a$$

What does **ambiguity** mean in the context of this grammar?

- A) One string $\rightarrow$ one parse tree
- B) One string $\rightarrow$ multiple parse trees
- C) No derivation
- D) Infinite symbols

Answer

- B) One string $\rightarrow$ multiple parse trees

Given CFG

$$S \rightarrow SS \mid a$$

Explanation

A CFG is called **ambiguous** if there is at least one string in its language that has **more than one parse tree**.

For the given grammar:

$$S \rightarrow SS \mid a$$

the production:

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$S \rightarrow SS$

allows a string to be divided into different groups of S.

For example, consider:

aaa

It can be grouped as:

(aa)a

or:

a(aa)

These different groupings produce different parse trees.

Derivation 1

S

$\rightarrow SS$

$\rightarrow SSS$

$\rightarrow aaS$

$\rightarrow aaa$

Derivation 2

S

$\rightarrow SS$

$\rightarrow aS$

$\rightarrow aSS$

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→ aaa

Thus, the same string can have more than one valid derivation structure.

Therefore:

One string → Multiple parse trees

Final Answer

B) One string → multiple parse trees

QUESTION 9

Question

Given the grammar:

$S \rightarrow 0S1 \mid 01$

Which of the following derivations correctly generates the string 0011?

A) $S \rightarrow 0S1 \rightarrow 00S11 \rightarrow 0011$

B) $S \rightarrow 01 \rightarrow 0011$

C) $S \rightarrow 0S1 \rightarrow 01$

D) $S \rightarrow 00S11 \rightarrow 0011$

Answer

A) $S \rightarrow 0S1 \rightarrow 00S11 \rightarrow 0011$

Given CFG

$S \rightarrow 0S1 \mid 01$

Step-by-Step Derivation

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Start with:

S

Apply:

$S \rightarrow 0S1$

Therefore:

$S \rightarrow 0S1$

Apply the same production again:

$0S1 \rightarrow 00S11$

Now replace S using:

$S \rightarrow 01$

Therefore:

$00S11 \rightarrow 00(01)11$

Simplifying:

$00(01)11 \rightarrow 0011$

Complete Derivation

S

$\rightarrow 0S1$

$\rightarrow 00S11$

$\rightarrow 00(01)11$

$\rightarrow 0011$

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Explanation

The production:

$$S \rightarrow 0S1$$

adds one 0 at the beginning and one 1 at the end.

The production:

$$S \rightarrow 01$$

terminates the derivation.

Thus, the required string 0011 is obtained correctly.

Final Answer

$$A) S \rightarrow 0S1 \rightarrow 00S11 \rightarrow 0011$$

QUESTION 10

Question

Given the grammar:

$$S \rightarrow aS \mid Sb \mid ab$$

Which of the following strings can be generated by the grammar?

A) ab

B) aab

C) abb

D) All of the above

Answer

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D) All of the above

Given CFG

$$S \rightarrow aS$$

$$S \rightarrow Sb$$

$$S \rightarrow ab$$

The production

$$S \rightarrow ab$$

provides the basic string.

The productions

$$S \rightarrow aS$$

$$S \rightarrow Sb$$

can add more as to the beginning and more bs to the end.

Derivation for ab

Use the terminating production:

S

$\rightarrow ab$

Therefore:

$ab \in L$

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Derivation for aab

Start with:

S

$\rightarrow aS$

Now replace S with ab:

aS

$\rightarrow a(ab)$

$\rightarrow aab$

Complete derivation:

S

$\rightarrow aS$

$\rightarrow a(ab)$

$\rightarrow aab$

Therefore:

$aab \in L$

Derivation for abb

Start with:

S

$\rightarrow Sb$

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Now replace S with ab:

Sb

$\rightarrow (ab)b$

$\rightarrow abb$

Complete derivation:

S

$\rightarrow Sb$

$\rightarrow (ab)b$

$\rightarrow abb$

Therefore:

$abb \in L$

Explanation

The grammar first generates the basic string:

ab

Using:

$S \rightarrow aS$

additional as can be added to the beginning.

Using:

$S \rightarrow Sb$

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additional bs can be added to the end.

Therefore, the grammar can generate all three given strings:

ab

aab

abb

Final Answer

D) All of the above